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"DON'T SHAKE THE BABY": THE EFFECTIVENESS OF A PREVENTION PROGRAM

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Abstract—The suggestion has been made that a systematic approach to informing all parents of newborns about the dangers of shaking a baby is needed. The purpose of this study was to determine whether an educational campaign about the dangers of shaking a baby could influence parental knowledge, and would be perceived as helpful by new parents. A total of 15,708 parents of newborns in one urban county received a "Don't Shake the Baby" packet over a one year period. Moms ranged in age from 14 to 44 years, and 90% were white. A response postcard in the educational packet was returned by 3,293 parents (21%). More than three-fourths of respondents said the information was helpful to them, and 49% indicated that subsequent to reading the materials, they were less likely to shake their babies. Ninety-one percent said they thought other parents of newborns should read the material.

Key Words—Shaken baby syndrome, Colic, Antishaking campaign, Prevention.

INTRODUCTION

CHILD ABUSE RESULTS in substantial numbers of deaths and disability among small children (Caffey, 1972, 1974; Curran, 1984; Ludwig & Warman, 1984). Head trauma is the most frequent cause of permanent damage or death among abused infants and children, and shaking, or whiplashing, accounts for a significant number of those cases (Billmire & Myers, 1985; Krugman, 1985). In one retrospective study of fatal child abuse cases over a 20-year period, shaking was implicated as the cause of death in 13% of cases (Showers, Apolo, Thomas, & Beavers, 1985). Data from the Ohio Department of Health for 1986–1988 indicated that 15% of deaths due to child battering or other maltreatment (age < 5 years) were confirmed cases of shaken baby syndrome (SBS). An additional 11% of the cases were questionable as due to SBS, suggesting the possibility that more than one-fourth of preschoolers who died in this 3-year period were shaken to death. Some infants who were shaken to unconsciousness but recovered might not have been seen in a health care setting. Thus, the association between shaking and subsequent brain damage or delays in learning in these cases may never be established. It is possible, then, that the incidence of SBS may be grossly underreported.

Infants and small children are very vulnerable to head trauma when they are shaken. The combination of a heavy head, weak neck muscles, soft and rapidly growing brain, thin skull wall, and lack of mobility and control of the head and neck make them extremely vulnerable to injury from shaking (Caffey, 1972; Calder, Hill, & Scholtz, 1984; Kravitz, Diessen, Gom-

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berg, & Koroch, 1969; Ludwig & Warman, 1984; Thyen, September 1988). Mortality from hard shaking is high (Caffey, 1972; Curran, 1984; Horsham, McDonald, & Reid, September 1988; Krugman, 1985; Ludwig & Warman, 1984; Showers et al., 1985), and morbidity can include brain damage, partial or total blindness, and/or mental retardation (Caffey, 1972; Curran, 1984; Eagan, Whelan-Williams, & Brooks, 1985; Frank, Zimmerman, & Leeds, 1985; Horsham et al., September 1988; Krugman, 1985; Ludwig & Warman, 1984).

As early as 1972, Dr. John Caffey (1972), who first described shaken baby syndrome, recommended a massive public educational program on the dangers of shaking infants. That recommendation has yet to be carried out in a comprehensive and systematic way (Dykes, 1986), although numerous researchers and physicians since then have supported the recommendation (Caffey, 1974; Curran, 1984; Eagan et al., 1985; Ludwig & Warman, 1984; Showers et al., 1985; Showers & Apolo, 1986; Thyen, September 1988).

In addition to the documented number of cases of SBS deaths since Caffey made his recommendation, two factors speak to the current timeliness of an antishaking baby campaign. First, in at least one medical setting in the United States, parents have actually been encouraged to shake their babies when infants were experiencing apneic episodes (Chadwick, April 1988). By 1988, a Task Force on Child Abuse and Neglect became aware that in at least two cases in which these instructions were provided to parents, shaking resulted in brain damage to their infants. The California Medical Association has since issued a position paper stating that instructions to parents not to shake their babies should be disseminated.

A second factor which speaks to the need for an antishaking baby initiative is the rising number of drug-exposed infants. Chasnoff, a leading physician and researcher with these infants, has said that these babies spend "a lot of time crying" and are very difficult to soothe (Revkin, September 1989). The behavior most frequently mentioned in the literature as triggering hard shaking is, in fact, colic, or inconsolable crying (Caffey, 1974; Curran, 1984; Frank et al., 1985; Krugman, 1985; Ludwig & Warman, 1984).

BACKGROUND RESEARCH

In spite of the documented dangers of shaking infants, five studies conducted in the past eight years have revealed that many adolescents and adults do not recognize the dangers of shaking a child. In separate studies, the following question was administered to college students, high school students, adults, pregnant and parenting teenagers, and female inmates (Showers, 1989, 1990; Showers & Johnson, 1984, 1985):

Which statement is true about shaking a baby who cries a lot?

- Shaking helps parents blow off steam and really isn't harmful
- Shaking is better and safer than spanking a baby
- Shaking can cause brain damage or kill a baby
- Shaking is an okay way to teach kids that adults are stronger than they are
- I really don't know.

Results of the studies are summarized below:

1. In a 1982 study of 299 college students, slightly more than half (54%) recognized that shaking can cause brain damage or death.
2. A 1983 study of 676 urban high school students revealed a 50% recognition rate of the dangers of shaking a baby.
3. In a 1987 study, 117 parents bringing children to ten pediatric practices answered the

question listed above as part of an education/research project. The overall rate of correct response to the question prior to reading project materials was 74%.

4. In the fall of 1988, the question was given to 1,843 pregnant and parenting teenagers enrolled in junior high schools, high schools and vocational schools throughout Ohio. The rate of correct response prior to receiving any educational material was 67%.
5. In the spring of 1990, 1,607 females incarcerated at the Ohio Reformatory for Women responded to the question about the dangers of shaking babies, with an accuracy rate of 69%.

The findings from all five studies indicated that the level of knowledge among the general public about the dangers of shaking infants was inadequate. Even by the most conservative estimates, these findings suggested that at least 25%, and as many as 50% of parents or prospective parents did not know that shaking an infant was dangerous.

After baseline data were collected in the latter three studies described above, subjects were given a 5″ × 8″ card entitled "Crying: What Should I Do?" to read (see Figure 1). The information on the card addressed the normalcy of infant crying, ways to cope with colic, and reasons not to shake a baby. In these studies, the above mentioned question was readministered one week to five months after subjects read the card. Post-test averages by group on the question about shaking a baby ranged from 93–97%, representing significant improvements in knowledge for all groups after reading the "Crying Card." These data clearly indicate that parents can be educated regarding the dangers of shaking infants, and those findings served as a foundation for conducting the present project.

METHODOLOGY

The goal of the "Don't Shake the Baby" project was to increase the knowledge of parents of newborns about the dangers of shaking babies, and subsequently, to decrease the practice of shaking babies and the resulting incidence of blindness, disability and death. A concomitant objective of the project was to educate parents that crying in infants, a behavior which frequently precipitates shaking, is normal, and to provide positive information about how to cope with a colicky baby. The target population consisted of all parents of newborns in Franklin County, Ohio, from August 1989 through July 1990.

Project Procedure

In January 1989, letters were mailed to the chief executive officers and the obstetrics supervisors at six hospitals in the target county to inform them about the proposed "Don't Shake the Baby" project. The six hospitals represent all medical facilities in the county which have licensed maternity units. Representatives of all six hospitals agreed to a subsequent face-to-face meeting about the project, and all ultimately provided their formal support and collaboration.

In August 1989, the author conducted an in-service meeting with maternity supervisors and staff at each hospital. Orientation consisted of a brief background and need for the project, possible approaches to distributing the information during the project year, establishing a protocol suitable for use at individual sites, and giving each staff member a series of Child Behavior Management Cards as a thank you for their assistance with the project.

The procedure for disseminating project materials at most hospitals consisted of giving the packet to the new mother when the birth certificate information was collected, and suggesting she read the material. This protocol was chosen because mothers tend to be very attentive to the completion of the birth certificate, and nursing staff felt that information given at that

No one likes to listen to a baby cry for a long time. It is irritating and frustrating. Some babies cry a lot when they are hungry, wet, tired, or just want company. Some infants cry at certain times of the day or night (usually when you want to sleep or eat.) Feeding and changing them may help, but sometimes even that doesn't work.

You may want to take your baby to the doctor to see if a medical reason can be found. Some babies cry because they have "colic." These babies seem to have a hard time settling in to life. Their crying may just be the way they deal with tense feelings. Often you can tell a "colicky" baby because s/he curls up, then straightens out over and over. →

If your baby cries a lot, try the following:

- Feed the baby slowly; burp the baby often.
- Offer the baby a pacifier.
- Hold the baby against your chest and walk or rock him/her.
- Take the baby for a ride in a stroller or car or put him/her in a baby swing.
- If you breastfeed, avoid eating onions or beans or drinking coffee, tea, or cola.

Be patient. The baby does not hate you or want to ruin your life. If you have had all you can take, wrap the baby in a soft blanket and put the baby on his/her stomach or right side (in a quiet, dark room if possible). Take a break. Have someone else take care of the baby for a while if possible.

No matter how impatient or angry you feel, DO NOT SHAKE THE BABY. (Hard shaking can cause brain damage or death.) Letting the baby cry it out when you have had enough is safer than shaking or punishing the baby. Your baby will outgrow the constant crying. For now, HOLDING AND CUDDLING BEHAVIOR TELLS THE BABY YOU LOVE HIM OR HER AND WANT HIM OR HER TO FEEL BETTER.

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Figure 1. Crying Card.

CONGRATULATIONS

ON YOUR NEW BABY!

You will be the most important person to this baby for a long time. The information in this packet can help you get off to a better start.

WILL YOU PLEASE TAKE A FEW MINUTES TO READ THIS "Crying: What Should I Do?" card, answer the questions on the postcard, and drop the postcard in the mail?

Thousands of new parents are receiving this information. YOUR opinion will help us decide if we should keep giving it to parents.

Thanks For Your Help

Figure 2. Project envelope.

time would be perceived as important. In one hospital, the packet was given to mothers with other information and baby products upon their release from the hospital.

Project Materials

The materials for the project consisted of a 6″ × 9″ envelope congratulating parents on their newborns and encouraging them to read the material provided (see Figure 2), a "Crying: What Should I Do" Card, and a self-addressed stamped postcard for responding to the materials. The outer envelope and postcard were color coded by race (hospital staff distributed packets according to the race of the mother), and the postcard requested information about age, and specific responses to the informational packet.

Table 1. Percentage of Total Respondents Answering "True" To Each Question (N = 3293)

I "did not" read the Crying Card	1%
I "did" read the Crying Card	99%
I found the card helpful	77%
I learned more about the danger of shaking a baby	57%
I am not as likely to shake my baby now	49%
I still don't think shaking a baby is dangerous	0%
I think other parents should read the Crying Card	91%

RESULTS

Demographic Variables

According to monthly statistics provided by participating hospitals, a total of 15,708 births occurred during the period of the project. Response postcards were received from 3,293 parents, representing a 21% return rate. Return rates ranged from 4% at one hospital to 35% at another hospital. The hospital with the lowest return rate was the site at which packets were distributed with other materials upon the mother's release. At the hospital with the highest postcard return rate, the attending nurse not only encouraged the mother to read the material and to complete the postcard, but also offered to mail the postcard from the nurses' station.

Although the packets were intended for reading by both mothers and fathers, in the vast majority of cases (95%), it was the mother who completed the postcard. Postcard respondents ranged in age from 14–44 years, with a mean of 27 years. For parents for whom race was known, 90% were white and 10% were nonwhite.

Parental Responses

Table 1 summarizes the responses to specific items on the postcard. In addition to the items shown in the table, the multiple-choice question described earlier regarding the effect of shaking a baby was also asked. Ninety-eight percent of respondents selected the correct response, "Shaking can cause brain damage or kill a baby."

Findings by Race

For five of the six hospitals which participated in this study, data were available by race (n = 2683). Findings are summarized in Table 2. Although project materials were well received by both white and nonwhite parents, minority parents had consistently higher rates of favor-

Table 2. Responses by Race

	White		Minorities	
	N	%	N	%
I "did not" read the Crying Card	19	1	2	1
I "did" read the Crying Card	2381	99	271	99
I found the card helpful	1816	75	238	87
I learned more about the danger of shaking a baby	1309	54	193	70
I am not as likely to shake my baby now	1114	46	168	61
I still don't think shaking a baby is dangerous	2	0	0	0
I think other parents should read the Crying Card	1860	90	216	93

able feedback. For instance, 86% of nonwhite parents said the information was helpful, compared to 75% of white respondents. Affirmative responses to learning more about the dangers of shaking were 70% for nonwhites vs. 54% for whites, and 61% of minority parents versus 46% of white parents said they are now less likely to shake their babies.

DISCUSSION

Little research has been conducted in the area of child abuse prevention programming to determine whether parents read available information and/or whether they find it helpful. Although there was no way to know whether the parents in this project who did not return the postcard read the material and/or benefitted by it, it was encouraging that more than one-fifth of parents receiving the project materials read them and responded by mail.

In the study, more than three-fourths of respondents said the information was helpful, 57% learned more about the dangers of shaking a baby, and almost half were less likely to shake their babies after reading the Crying Card. These findings support the merit of this educational initiative. It is possible that the 51% who did not indicate on the return postcard that they were less likely to shake their babies were already informed about the dangers of shaking and would not have shaken a baby prior to reading the materials. Previous research findings support this possibility (Showers & Johnson, 1984, 1985; Showers, 1989, 1990). Perhaps the most poignant finding was that 91% of respondents thought other parents should read these materials.

Although it is not possible to prove whether improved knowledge and prediction of nonshaking responses translate into actual behavior, the previously documented association between lack of knowledge and abusive behavior suggests that this educational campaign is worthwhile. An ongoing review of specific causes of child abuse fatalities will assist in determining whether the incidence of deaths due to SBS decreases as this educational campaign expands.

Since materials in this project were delivered directly to mothers, it was not surprising that most of the follow-up data were received from moms. More effort must be made to distribute the information to male caretakers since many SBS incidents are perpetrated by men.

In order to further evaluate the cultural appropriateness of the Don't Shake the Baby model, more data are needed from nonwhite populations. Expansion of the current project to a large, urban, minority population is underway. However, the findings by race in the present study suggest preliminarily that the Don't Shake the Baby model is as effective, if not more effective, for nonwhites as for whites.

CONCLUSIONS

The findings in the study were limited to 3,293 of 15,708 parents of newborns in one urban county who received an informational packet about the dangers of shaking a baby. The sample represents only those parents who voluntarily responded to project materials. Results from the study suggest the following conclusions and recommendations: (a) A need exists to educate parents about the dangers of shaking a young child; (b) parents perceive information about the dangers of shaking, and about how to manage colic, as helpful; (c) the Don't Shake the Baby model is an effective model for raising parental knowledge about the dangers of shaking an infant; (d) the model tested has potential benefit to parents of diverse cultural backgrounds; and (e) the model used in this study is worthy of replication and further study.

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Résumé—Il a été suggéré qu'une information systématique de tous les parents de nouveau-nés concernant les dangers liés au fait de les secouer était nécessaire. Le but de cette étude a été d'évaluer si une campagne éducative concernant ce danger pouvait influencer le savoir des parents et serait perçue par eux comme utile. Un total de 15.708 parents de nouveaunés dans une région urbaine a pu bénéficié de la panoplie "Ne secouer pas votre bébé" pour une période d'un an. L'âge des méres variait entre 14 et 40 ans et 90 pourcent d'entre elles étaient blanches. Une carte postale comprise dans la panoplie et destinée á répondre a été renvoyée par 3.293 parents (21 pourcent). Plus de trois quart des répondants affirmaient que l'information leur avait été utile et 49 pourcent d'entre eux signalaient qu'après avoir lu le matériel envoyé, ils seraient moins enclins á secouer leurs bébés. Nonante et un pourcent pensaient aue d'autres parents de nouveau-nés devraient lire ce matériel.

Resumen—Se ha sugerido que es necesario informar a todos los padres de recién nacidos de los peligros al sacudir un bebé. El objetivo de este trabajo fue determinar si una campaña educativa sobre los peligros de sacudir un bebé podía influir en los conocimientos de los padres y sería percibida como útil por los nuevos papás. Un total de 15,708 padres de recién nacidos recibieron una carpeta con "No Sacuda al Bebé!" por un período de un año. Las mamás variaban en edad desde 14 a 44 años y el 90% eran blancas. Unos 3,293 padres (21%) devolvieron la tarjeta de respuesta incluída en la carpeta. Más de 3/4 partes de los que respondieron dijeron que la información les había sido de utilidad y 49% indicaron que después de leer el material de la carpeta, tenían menos riesgo de sacudirle la cabeza a sus bebés. El 91% dijo que pensaban que otros padres de recién nacidos debían leer el material.